Report

Of

Topic

Cisco Virtual Internship Program 2026 By Cisco

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE

By

Roll Number of Student: 2410030019

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Section – 3CSE1

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Batch: B.TECH CSE AIML

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CANDIDATE'S DECLARATION

I, **AYAN MANDAL** do hereby declare that the internship report titled

"Cisco Virtual Internship Program 2026 By Cisco"

has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:

Student Name - **AYAN MANDAL**

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ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluably constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.
I express my gratitude to my parents for their blessings.

Signature

Student Name – **AYAN MANDAL**

Roll No - **2410030019**

<u>Instruction for the report format</u>

- Cover Page, Copy of the Certificate issued by industry/organisation will have no Page Number.

- Page no. of Sr. No. 1 to 6 along with a list of tables and list of figures will bear the Roman Numbering System (e.g. i, ii, iii).
- Bibliography/References should be in proper format.

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Tables, Figures and Graphs:	With Numbers indicating the chapters (1.1, 1.2 or 2.1, 2.2 etc.) and contain titles (above for tables

4. PROJECT DESCRIPTION

4.1 Introduction

The **Cisco–AICTE Virtual Internship Program 2026** provided an opportunity to gain practical knowledge of modern technologies through Cisco Networking Academy. The internship focused on developing an understanding of **Artificial Intelligence (AI), AI applications, and Data Analytics**.

As part of the learning journey, three major Cisco courses were explored: **Introduction to Modern AI**, **Apply AI: Analyze Customer Reviews**, and **Data Analytics Essentials**. These courses provided knowledge ranging from fundamental AI concepts to the practical application of AI for analyzing customer feedback and the use of data analytics techniques for generating meaningful insights.

The **Introduction to Modern AI** course introduced fundamental concepts such as Artificial Intelligence, Machine Learning, Generative AI, Computer Vision, Chatbots, Machine Translation, and Prompting. It helped establish a basic understanding of how modern AI technologies are being used in different real-world applications.

The **Apply AI: Analyze Customer Reviews** course focused on applying AI techniques to unstructured customer feedback. It introduced concepts such as topic identification, thematic analysis, sentiment analysis, translation, data cleaning, and human verification. These techniques demonstrate how large amounts of customer feedback can be converted into structured information.

The **Data Analytics Essentials** course focused on the fundamentals of data analytics and the process of transforming raw data into useful insights. It introduced concepts related to data preparation, analysis, visualization, dashboards, and data-driven decision-making.

Overall, the internship helped develop an understanding of how **AI and Data Analytics can work together to solve practical problems**. The learning experience also improved analytical thinking, problem-solving, technical awareness, and understanding of modern industry technologies.

4.2 Organization Profile

Cisco

Cisco is a global technology company that develops networking, security, collaboration, and other technology solutions. Cisco also works extensively in technology education through its **Cisco Networking Academy**, which provides learners with opportunities to develop technical and industry-relevant skills.

Cisco Networking Academy offers learning programs in areas including networking, cybersecurity, programming, artificial intelligence, data analytics, and other emerging technologies. The platform combines theoretical learning with practical activities, assessments, and skill development.

Cisco Networking Academy

Cisco Networking Academy is an educational platform designed to help students and professionals develop technology skills. It provides structured courses, learning modules, practical activities, assessments, and digital credentials.

Through its courses, learners can develop both foundational knowledge and practical understanding of technologies used in the modern IT industry.

Cisco–AICTE Virtual Internship

The Cisco–AICTE Virtual Internship Program provides students with an opportunity to learn industry-relevant technologies through a virtual learning environment. The program connects academic learning with practical technology skills and encourages students to develop knowledge that can be applied to real-world problems.

The internship learning in this report focuses on the areas of **Artificial Intelligence and Data Analytics**, with particular emphasis on understanding AI applications, customer review analysis, and data-driven decision-making.

4.3 Problem Statement

Organizations generate large amounts of data from customers, applications, websites, social media, transactions, and other sources. A significant portion of this information is available in an unstructured form, such as customer reviews and feedback.

Manually processing a large number of customer reviews can be time-consuming and difficult. It may also be challenging to identify common topics, customer concerns, positive feedback, and overall sentiment.

Traditional data analysis methods may also require considerable effort to transform raw information into meaningful insights. Therefore, there is a need for efficient methods that combine **Artificial Intelligence and Data Analytics** to process information and support decision-making.

The problem addressed through this learning project can therefore be stated as:

How can Artificial Intelligence and Data Analytics techniques be used to analyze customer feedback, identify important patterns and sentiments, and convert the results into meaningful insights for better decision-making?

The project explores how AI-assisted analysis and data analytics can be combined to address this problem.

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4. PROJECT DESCRIPTION

4.4 Project Objectives

The major objectives of the project are:

1. To understand the fundamental concepts of Artificial Intelligence and modern AI technologies.
2. To study the basic concepts of Machine Learning and Generative AI.
3. To understand the applications of AI in real-world scenarios.
4. To learn how AI tools and prompting techniques can be used effectively.
5. To understand how customer reviews can be analyzed using AI-assisted techniques.
6. To identify important topics and themes from customer feedback.
7. To understand the concept of sentiment analysis and its application to customer reviews.
8. To understand the importance of data cleaning and preparation.
9. To learn fundamental Data Analytics concepts.
10. To understand how structured and unstructured data can be analyzed.
11. To learn about data visualization and dashboards.
12. To understand how analytical results can be communicated through data storytelling.
13. To develop analytical and problem-solving skills.
14. To understand how AI and Data Analytics can support data-driven business decisions.
15. To gain exposure to industry-relevant technology concepts through Cisco Networking Academy.

4.5 Scope of the Project

The scope of this project covers the application of Artificial Intelligence and Data Analytics to the analysis of customer feedback and generation of meaningful insights.

The project focuses on understanding how customer reviews can be processed, categorized, and analyzed using AI-assisted techniques. The analysis can help identify common topics, themes, positive and negative sentiments, and areas requiring improvement.

Major areas within the scope include:

Artificial Intelligence

- AI fundamentals
- Machine Learning concepts
- Generative AI
- Chatbots
- Computer Vision
- Prompting

Customer Review Analysis

- Data preparation
- Topic identification
- Thematic analysis
- Sentiment analysis
- Translation
- Duplicate identification
- Human verification

4.5 Scope of the Project (Cont.)

Data Analytics

- Data collection
- Data cleaning
- Data organization
- Data analysis
- Data visualization
- Dashboard development
- Data storytelling

Potential Applications

The concepts learned can be applied in:

- Customer experience management
- E-commerce
- Hospitality
- Food delivery
- Retail
- Banking
- Healthcare
- Education
- Social media analysis
- Business intelligence
- Marketing analytics

The project also provides a foundation for future development of more advanced systems using Machine Learning, Natural Language Processing, Generative AI, and predictive analytics.

4.6 Technologies and Tools Used

The internship introduced several technologies and tools related to AI and Data Analytics.

Artificial Intelligence

AI concepts were studied to understand how computer systems can perform tasks that normally require human intelligence, including pattern recognition, classification, generation, and decision support.

Generative AI

Generative AI was explored as a technology capable of generating new content and assisting users in solving different types of problems.

Large Language Models and Prompting

Prompting techniques were studied to understand how effective instructions can be provided to AI systems to obtain more useful and relevant responses.

Sentiment Analysis

Sentiment analysis was studied for identifying whether customer feedback expresses positive, negative, or mixed opinions.

Thematic Analysis

Thematic analysis was used conceptually to group related topics into broader themes and understand common patterns in customer feedback.

Excel / Spreadsheets

Spreadsheets can be used for organizing, cleaning, filtering, calculating, and analyzing tabular data.

4.6 Technologies and Tools Used (Cont.)

SQL

SQL is used to retrieve, filter, organize, and analyze information stored in relational databases.

Tableau

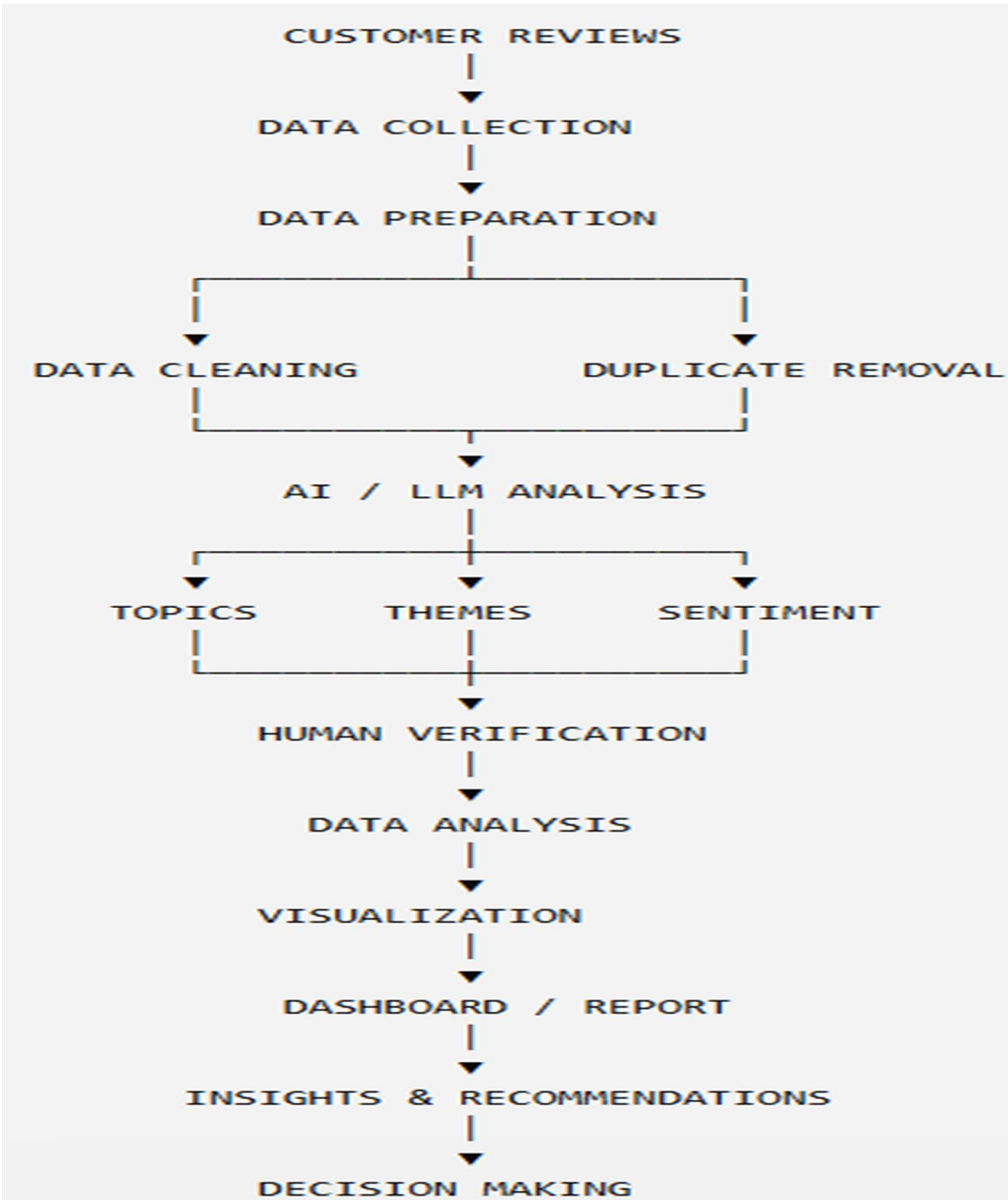
Tableau is used for creating visualizations and dashboards that help users understand patterns and trends in data.

Data Visualization

Data visualization presents analytical results through charts, graphs, and dashboards, making complex information easier to understand.

4.7 System Architecture

The proposed conceptual architecture combines AI-based customer review analysis with data analytics.



4.7 System Architecture (Cont.)

Architecture Explanation

The system begins with the collection of customer reviews. The collected information is prepared and cleaned before analysis.

AI-assisted techniques can then be used to identify topics, group related themes, and determine customer sentiment. The generated results should be reviewed and verified to ensure that the analysis is meaningful.

The processed information can then be analyzed using data analytics techniques and represented through charts, dashboards, or reports. Finally, the generated insights can support business decisions and identify areas for improvement.

4.8 Methodology

The project follows a systematic methodology consisting of several stages.

Step 1: Understanding the Problem

The first step is to understand the requirement for analyzing large amounts of customer feedback and identifying useful information from it.

Step 2: Data Collection

Customer reviews or feedback are collected from the available data source.

Step 3: Data Preparation

The collected information is organized into a suitable format for further processing.

Step 4: Data Cleaning

Unnecessary information, duplicate entries, and irrelevant content are identified and removed where appropriate.

Step 5: Topic Identification

AI-assisted techniques are used to identify the major topics discussed in customer reviews.

Step 6: Thematic Analysis

Related topics are grouped into broader themes such as product quality, customer service, delivery, price, and other relevant categories.

Step 7: Sentiment Analysis

Customer feedback is analyzed to identify positive, negative, or mixed sentiment.

Step 8: Human Verification

The AI-generated results are reviewed by a human to identify errors and improve the reliability of the analysis.

Step 9: Data Analysis

The processed information is analyzed to identify patterns, frequencies, trends, and important observations.

4.8 Methodology (Cont.)

Step 10: Data Visualization

The analytical results are presented using charts, graphs, or dashboards to make the findings easier to understand.

Step 11: Insight Generation

The results are interpreted to identify customer concerns, strengths, weaknesses, and improvement opportunities.

Step 12: Recommendations

Based on the findings, possible actions and recommendations are identified to support better decision-making.

4.9 Expected Outcomes

The project is expected to provide the following outcomes:

1. A clear understanding of modern Artificial Intelligence concepts.
2. Improved knowledge of Generative AI and AI-assisted applications.
3. Understanding of how prompting can be used to interact effectively with AI systems.
4. Knowledge of how customer reviews can be analyzed using AI-assisted techniques.
5. Ability to identify topics and themes from customer feedback.
6. Understanding of sentiment analysis and its practical applications.
7. Knowledge of data cleaning and preparation techniques.
8. Understanding of the complete data analytics workflow.
9. Improved knowledge of SQL and spreadsheet-based analysis.
10. Understanding of data visualization and dashboard concepts.
11. Ability to interpret analytical results and identify meaningful patterns.
12. Development of analytical and problem-solving abilities.
13. Better understanding of data-driven decision-making.
14. Improved awareness of the practical applications of AI and Data Analytics in industry.
15. A foundation for future projects in Artificial Intelligence, Machine Learning, Data Analytics, Business Intelligence, and Data Science.

4.10 Bibliography

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4.11 Certificates of Completion

1. Introduction to Modern AI

Cisco
Networking
Academy

IILM UNIVERSITY

This certificate is awarded to

Ayan Mandal

for successfully completing

Introduction to Modern AI

offered by IILM University
through the Cisco Networking Academy program.

Swati Vashisht
Instructor
IILM University

30 May 2026
Completion Date

Cert ID: 8845de6a-668a-4f39-9c83-2b586cdde961

4.11 Certificates of Completion

2. Apply AI: Analyze Customer Reviews

Cisco
Networking
Academy

IILM
UNIVERSITY

This certificate is awarded to

Ayan Mandal

for successfully completing

Apply AI: Analyze Customer Reviews

offered by IILM University
through the Cisco Networking Academy program.

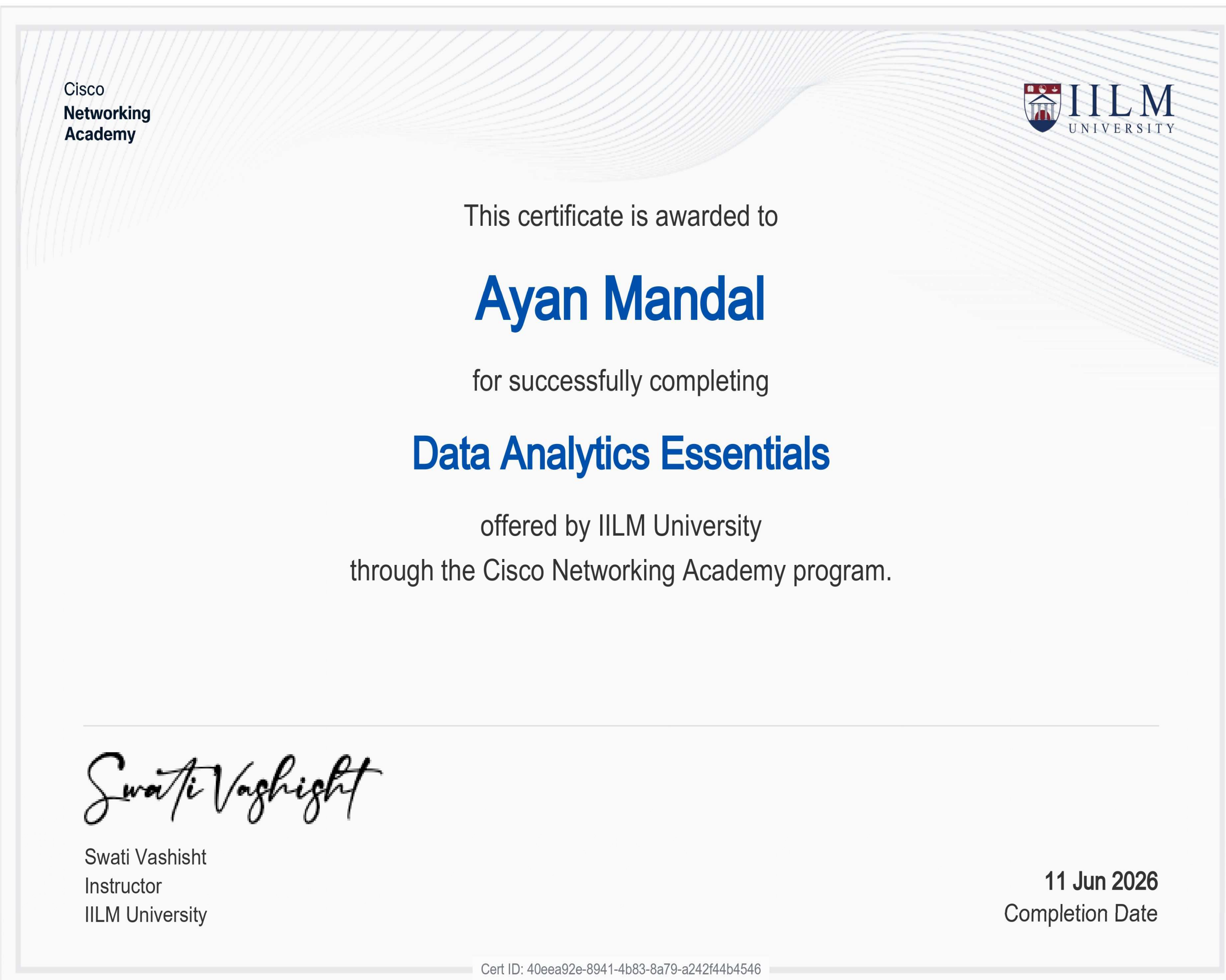
Swati Vashisht
Instructor
IILM University

30 May 2026
Completion Date

Cert ID: 48df18ac-bf7a-4025-9a58-55a229d005e9

4.11 Certificates of Completion

3. Data Analytics Essentials



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IILM University

11 Jun 2026
Completion Date

Cert ID: 40eea92e-8941-4b83-8a79-a242f44b4546